PROBABILITY & STATISTICS FINAL REVIEW 3

A Statistical Comparison of Human Written and AI Generated Code

Across C, C++, Java and Python

Code length
Normalized text density
Comment presence

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C low level language

C++ multi paradigm

Java object oriented

Python high level

10000 code samples
5000 AI & 5000 human

What Review 3 adds after the earlier reviews:

In review 1–2, we established raw differences. In review 3 we ask a more specific question: what changes when we normalize and model the data?

01 Normalize code density

We ccorrected “characters per nonblank line” uses only characters on nonblank lines and excludes CR/LF line breaks.

02 Split comment presence from volume

This tests whether a sample has any detected comment separately from how many comment lines were counted.

03 Model both factors together

Expanded regression and factorial ANOVA test language-by-origin interaction, not just separate averages.

04 Audit uncertainty

We corrected Cliff’s delta, regression diagnostics, corrected density output and manual comment vvalidation sample.

Review 3’s contribution is a more interpretable style comparison, not just another round of the same mean tests.

Research question and objectives

Do measurable code characteristics differ by origin, and does the pattern depend on programming language (we use java, python, C and C++).

01 **Size**

We compare line and character counts.

02 **Density**

Compare characters per nonblank line.

03 **Comments**

Separate comment presence from comment volume.

04 **Relationship**

Measure association and model character count.

05 **Interaction**

Test language and origin effects, even their effects together (language-by-origin).

Scope of the project is to compare group level statistical patterns. This project does not classify individual snippets or measure code quality. But it can be used for further projects that do that.

Dataset and operational measurements

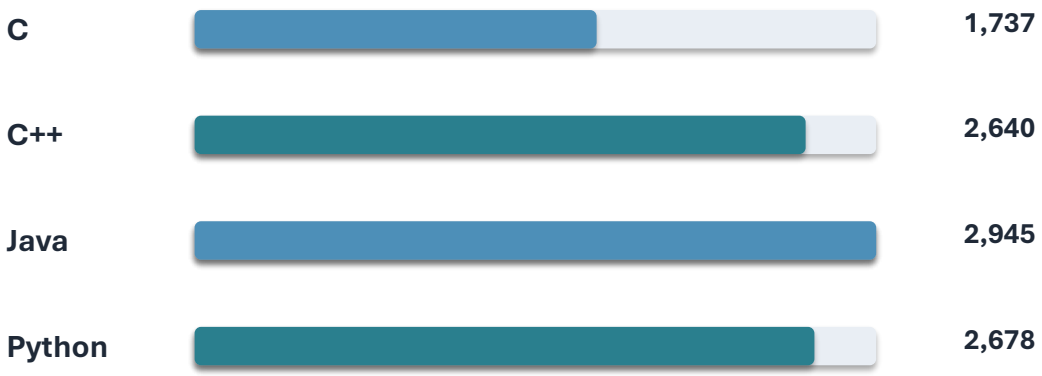
10000
code samples
Source dataset: Boukili et al. (2026)

5000
AI generated
Balanced overall group

5000
Human written
Balanced overall group

0
missing code values
0 exact duplicate full rows

Language distribution (all samples)



Please note that the 50/50 origin split is a dataset design feature here not really a real world prevalence.

Variables used:

Raw size

Line count and stored character count.

Normalized density

Characters on nonblank lines and number of nonblank lines.

Comment presence

At least one detected comment line in each code snippet.

Comment volume

Approximate count per 100 nonblank lines.

How all five PAS modules were integrated in this project:

MODULE 1	**Random variables**	Means, medians, SD, variance and dependence checks.	*What varies across samples?*
MODULE 2	**Distributions**	Histograms, outliers, skew and comment count overdispersion check.	*What do the distributions look like?*
MODULE 3	**Hypothesis testing**	Welch t-tests, confidence intervals, Holm correction, effect sizes, Wilcoxon and chi squared tests.	*Are the found differences supported by evidence?*
MODULE 4	**Correlation & regression**	Spearman ρ, log-linear regression and interaction comparisons.	*How do size measures relate?*
MODULE 5	**Factorial ANOVA**	Language, origin and language-by-origin on log transformed outcomes.	*Does the pattern vary by language?*

Please note that we applied the above as needed so not every syllabus topic or experimental design from each module is used, the listed ones are the ones we applied. Overall it's loosely based on our probability and statistics theory syllabus.

BASELINE RESULT (Explored in reviews 1 and 2)

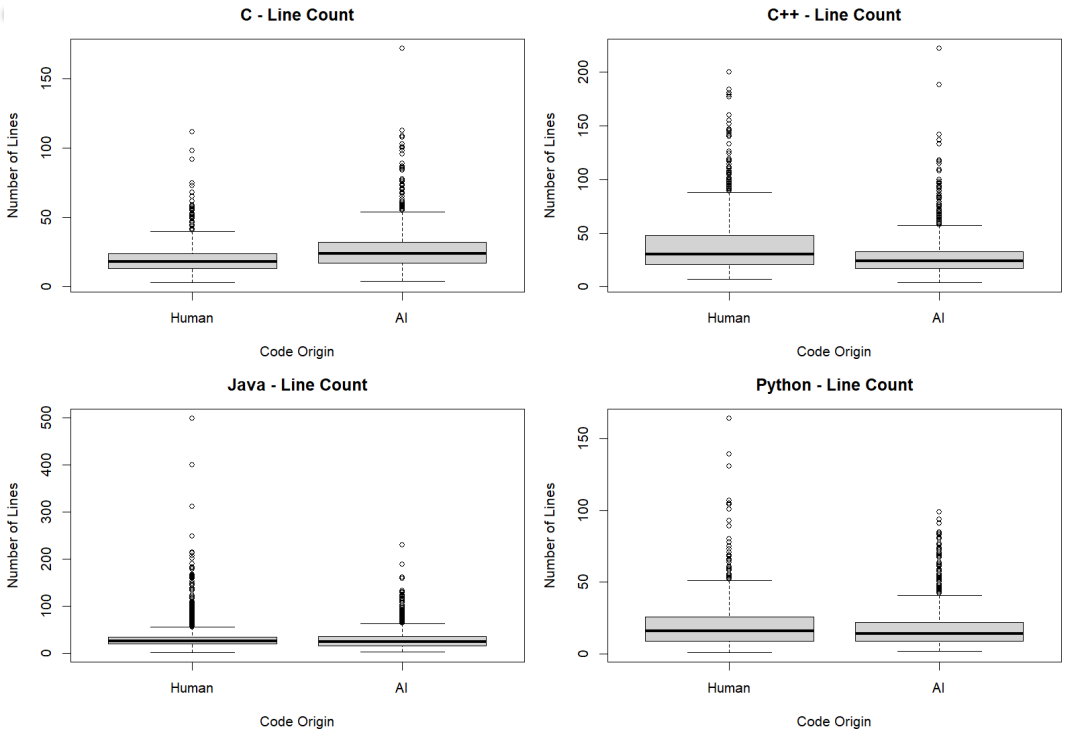
Let us start with the raw measurements

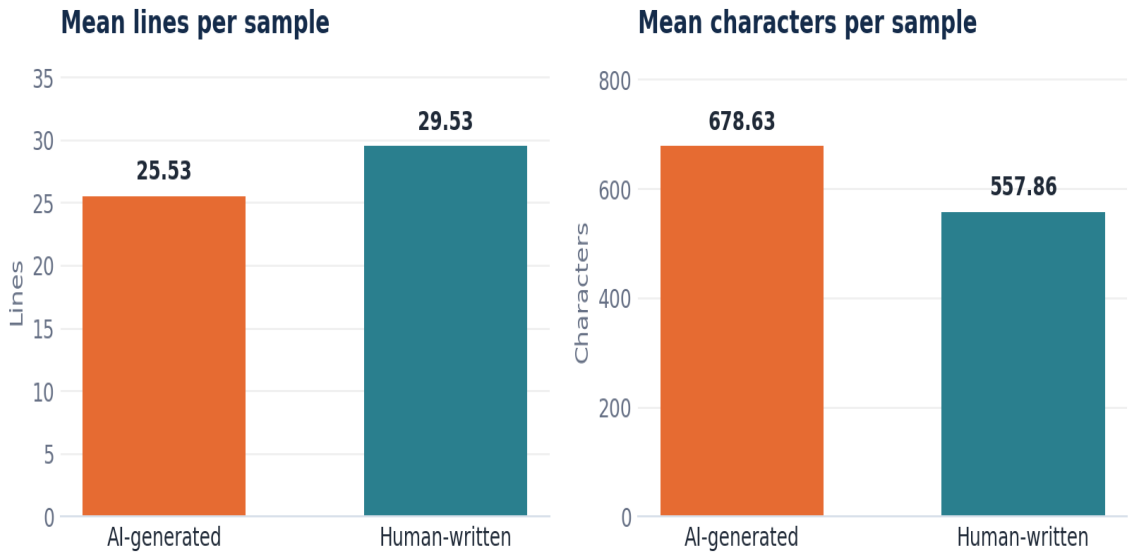
HUMAN-WRITTEN

Had +4.0 lines on average

AI-GENERATED

+120.8 characters on average



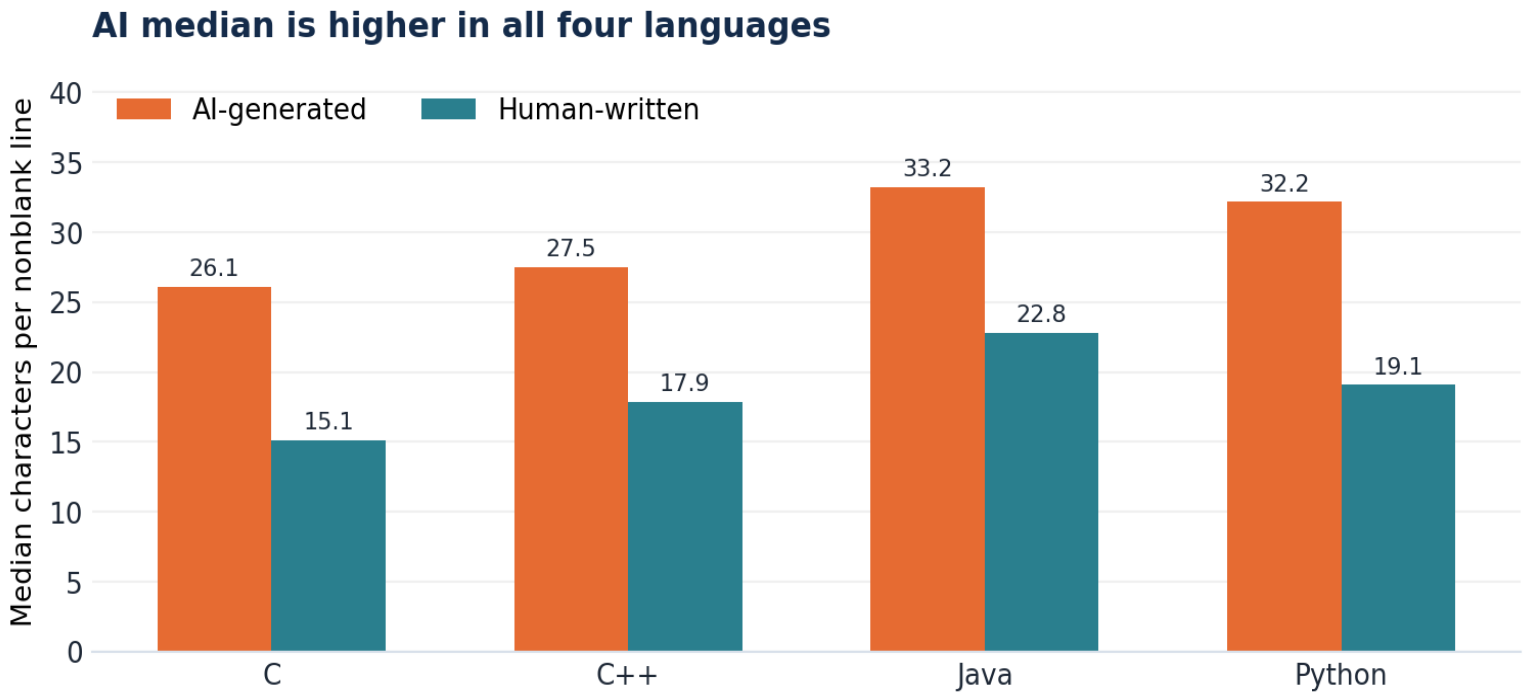


We can see that size alone is incomplete, line count and character count do not move in the same way, and the direction changes by language. So we applied the same tests to languages separately also.

We also noticed that the distributions are strongly right-skewed, with long upper tails and outliers, so we also analysed them based on median and rank.

NORMALIZED MEASURE

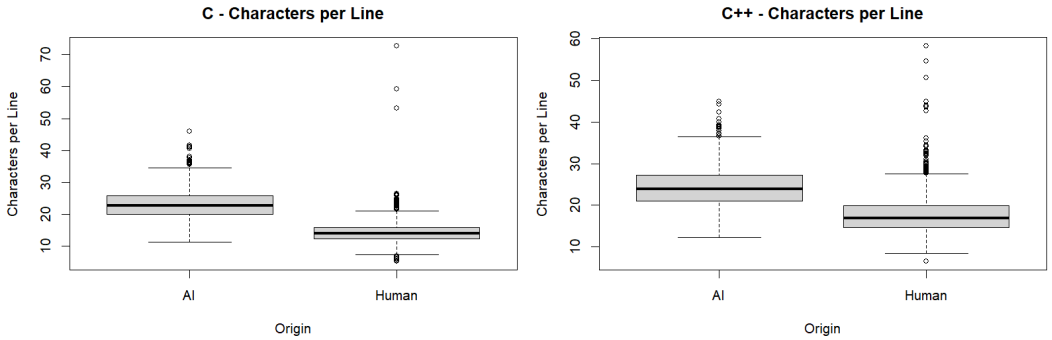
Finding 1: AI samples have higher text density

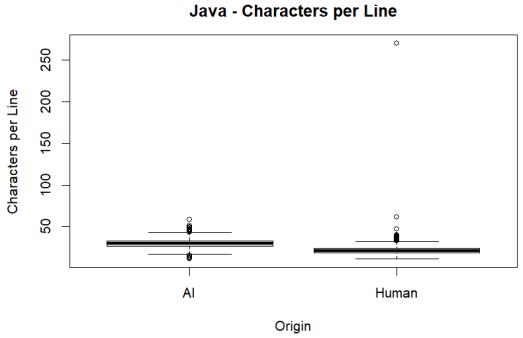


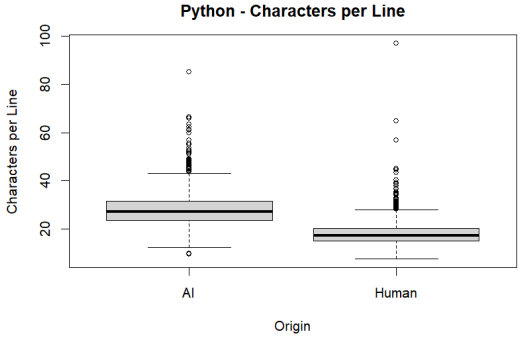
MEDIAN DIFFERENCE

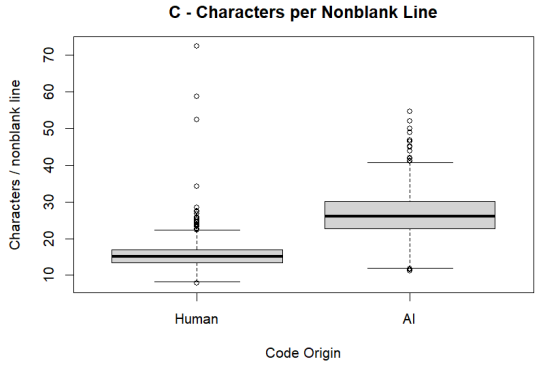
C	**+11.0 chars/line**
C++	**+9.6 chars/line**
Java	**+10.4 chars/line**
Python	**+13.1 chars/line**

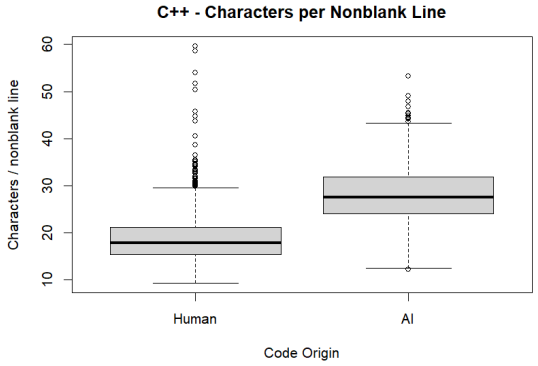
Definition: characters on nonblank source lines ÷ nonblank lines, excludes CR/LF and blank-line whitespace, but retains indentation, trailing spaces and such on nonblank lines. R graphs on the next slide.

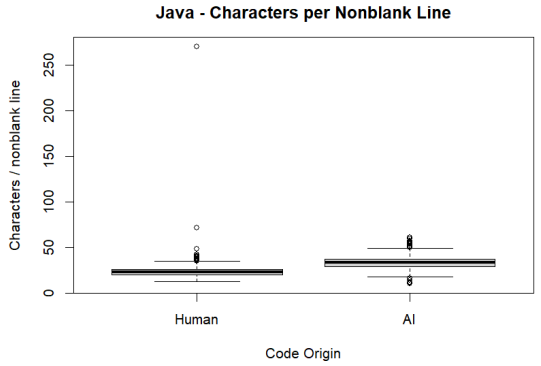


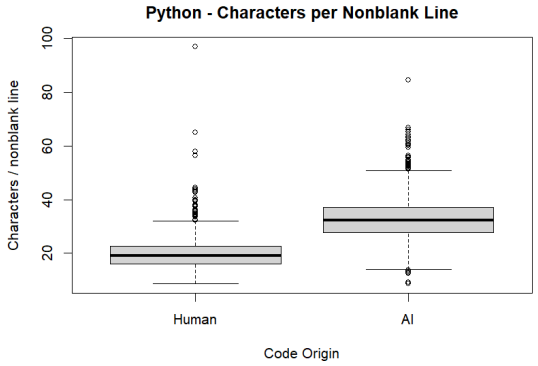






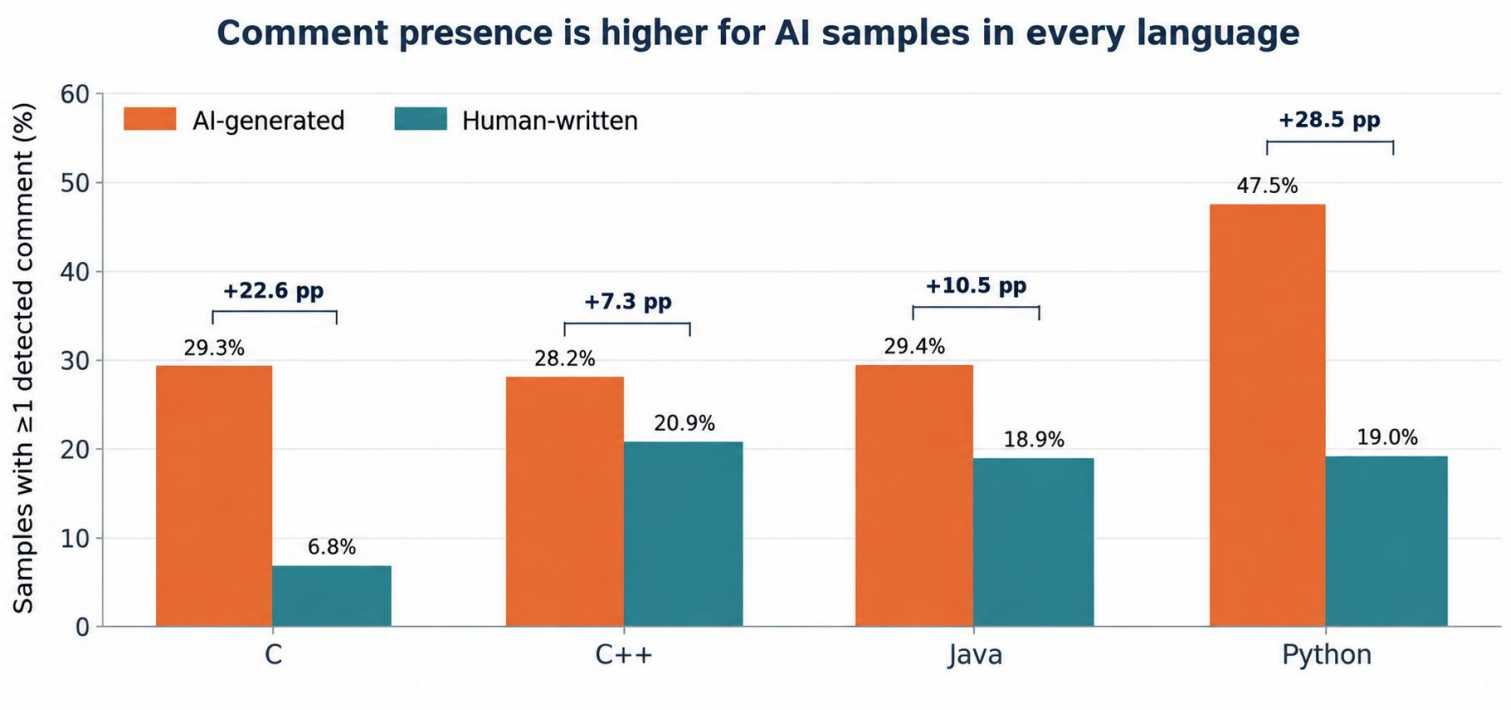






All 4 language comparisons remained significant after Holm correction, Cliff's delta was 0.80–0.92 basically means large rank separation.

Finding 2: Comment presence differs by language



LARGEST GAPS WERE:

Python	+28.5
C	+22.6
Java	+10.5
C++	+7.3

All 4 proportion comparisons remained significant after Holm correction (95% CIs exclude 0). Please note that this measures detectable comment presence, not quality of the comment. But since the #include and such comments will be common for both measures, the percentage difference can be trusted.

To note: comment volume did not move uniformly. You can see that human C++ samples had a higher mean counted comment count, all groups had median count 0 though.

Finding 3: For mean comparisons, we found strong evidence, with one important exception:

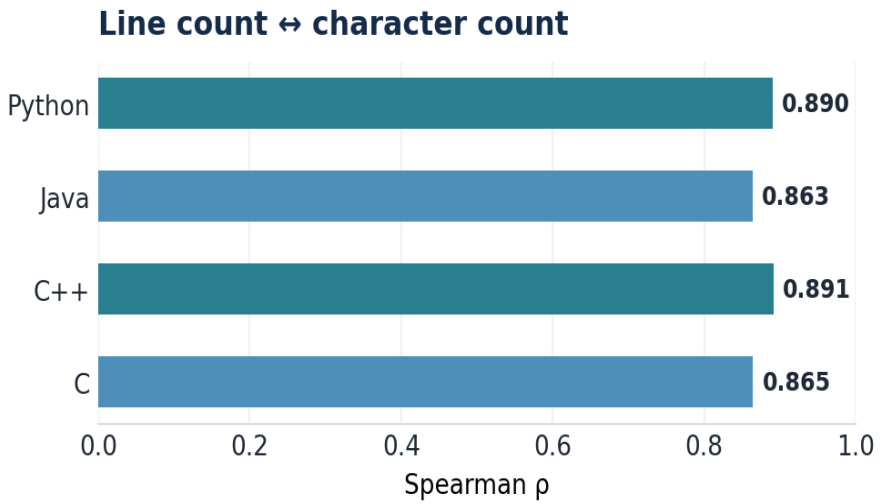
Language	Measure	Mean (AI/Human)	95% CI	Holm p (Welch)	Cliff’s Delta (rank)
C	Lines	+7.12	[5.76, 8.47]	<2.2e−16	+0.31
C	Characters	+359.60	[322.46, 396.75]	<2.2e−16	+0.69
C++	Lines	-9.15	[-10.89, -7.41]	<2.2e−16	-0.25
C++	Characters	+13.70	[-29.71, 57.11]	0.536	+0.07
Java	Lines	-4.48	[-6.40, -2.56]	1.4e-05	-0.11
Java	Characters	+149.15	[100.95, 197.35]	5.9e-09	+0.25
Python	Lines	-1.77	[-2.89, -0.65]	0.004	-0.09
Python	Characters	+143.78	[116.71, 170.86]	<2.2e−16	+0.27

Welch test: 7/8 were significant even after Holm correction. C++ character counts mean difference is not significant though (p = 0.536). It’s wilcoxon is significant but the rank effect is still small (Delta= +0.075).

Finding 4: Size measures are strongly associated

OVERALL SPEARMAN CORRELATION:

$\rho = 0.882$ AND $p < 2.2 \times 10^{-16}$



Consistently strong positive association within each language (ρ ≈ 0.863–0.891).

NESTED REGRESSION COMPARISONS:

Language X origin	**$F(3, 9991) = 63.83$** **$p = 7.08 \times 10^{-41}$**
Line slope X origin	**$F(1, 9990) = 10.07$** **$p = 0.00151$**

Adding language specific origin differences substantially improved model fit, so we did that. The line count slope interaction was also a smaller but statistically significant, improvement.

Regression models log(1 + character count). We actually added this in case anyone uses this study , such as training a model. In that case if the input character count is 0, log (1+0) will handle that error. Please also note that diagnostics show heavy residual tails and possible non constant variance; results are exploratory, not a validated predictor.

Finding 5: The origin pattern changes by language

Outcome:	Effect tested:	F statistic:	Partial η^2:	Holm adjusted p:
Lines	Language	532.1	0.138	<2.2e−16
Lines	Origin	47.9	0.005	4.6e-12
Lines	Language × origin	69.6	0.020	<2.2e−16
Characters	Language	601.6	0.153	<2.2e−16
Characters	Origin	544.5	0.052	<2.2e−16
Characters	Language × origin	105.2	0.031	<2.2e−16

Language effect:

This is the largest effect for both line and character counts.

Origin effect:

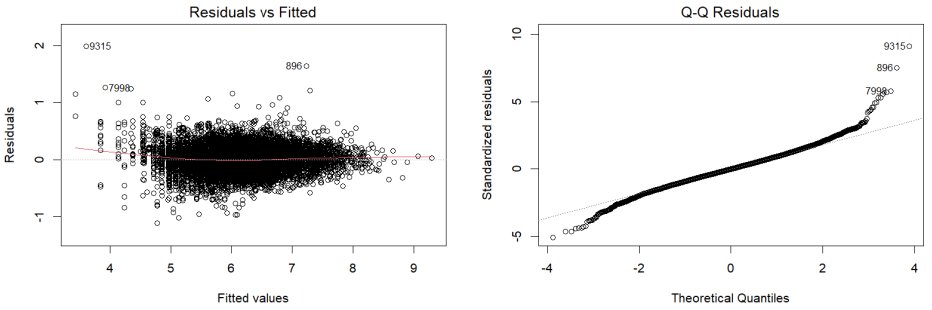
Small for line count (ηp^2 is about 0.005), larger for character count (about 0.052).

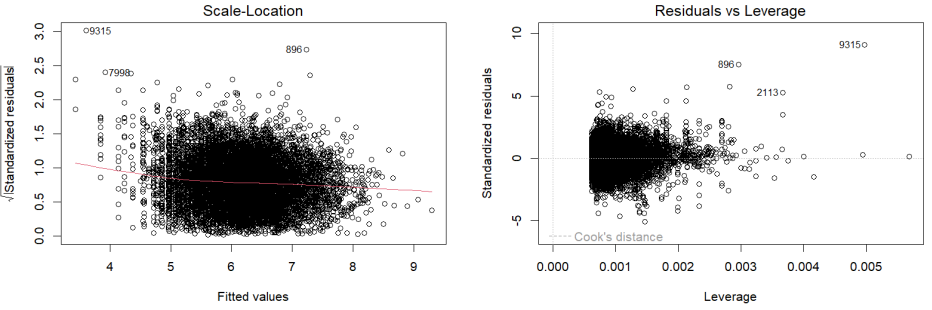
Interaction:

This is pretty significant for both outcomes: the origin associated pattern differs across languages.

A peek at the code, graphs, outputs, tables and the project directory:

Language	Origin	N	Mean	Variance	Variance_1	Observed_	Poisson_Expected_Zero_Probability
C	Human	769	0.248375	2.116615	8.52187	0.93238	0.780068
C	AI	968	0.669421	1.58864	2.373154	0.706612	0.512005
C++	Human	1419	0.940803	26.66504	28.34284	0.791402	0.390314
C++	AI	1221	0.57412	1.177494	2.050955	0.718264	0.563201
Java	Human	1605	0.730218	6.608594	9.050165	0.811215	0.481804
Java	AI	1340	0.718657	2.052975	2.856684	0.70597	0.487407
Python	Human	1207	0.584093	3.969498	6.796006	0.810273	0.557612
Python	AI	1471	0.995241	2.18161	2.192041	0.525493	0.369634





Final_Chi_Square_Output.txt

File Edit View H1

```
        Pearson's Chi-squared test

data:  origin_language_table
X-squared = 87.519, df = 3, p-value < 2.2e-16
```

```r
count_comments_heuristic<-function(code, language) {
  if (is.na(code) || !nzchar(code)) return(0L)
  lines <- strsplit(gsub("\r", "", code, fixed = TRUE), "\n", fixed = TRUE)[[1]]
  lines <- trimws(lines)
  total <- 0L
  in_block <- FALSE

  for (line in lines) {
    if (!nzchar(line)) next

    if (language == "Python") {
      if (grepl("^#", line)) total <- total + 1L
      next
    }

    if (in_block) {
      total <- total + 1L
      if (grepl("\\*/", line)) in_block <- FALSE
      next
    }

    if (grepl("^//", line)) {
      total <- total + 1L
      next
    }

    if (grepl("^/\\*", line)) {
      total <- total + 1L
      if (!grepl("\\*/", line)) in_block <- TRUE
    }
  }
  total
}
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

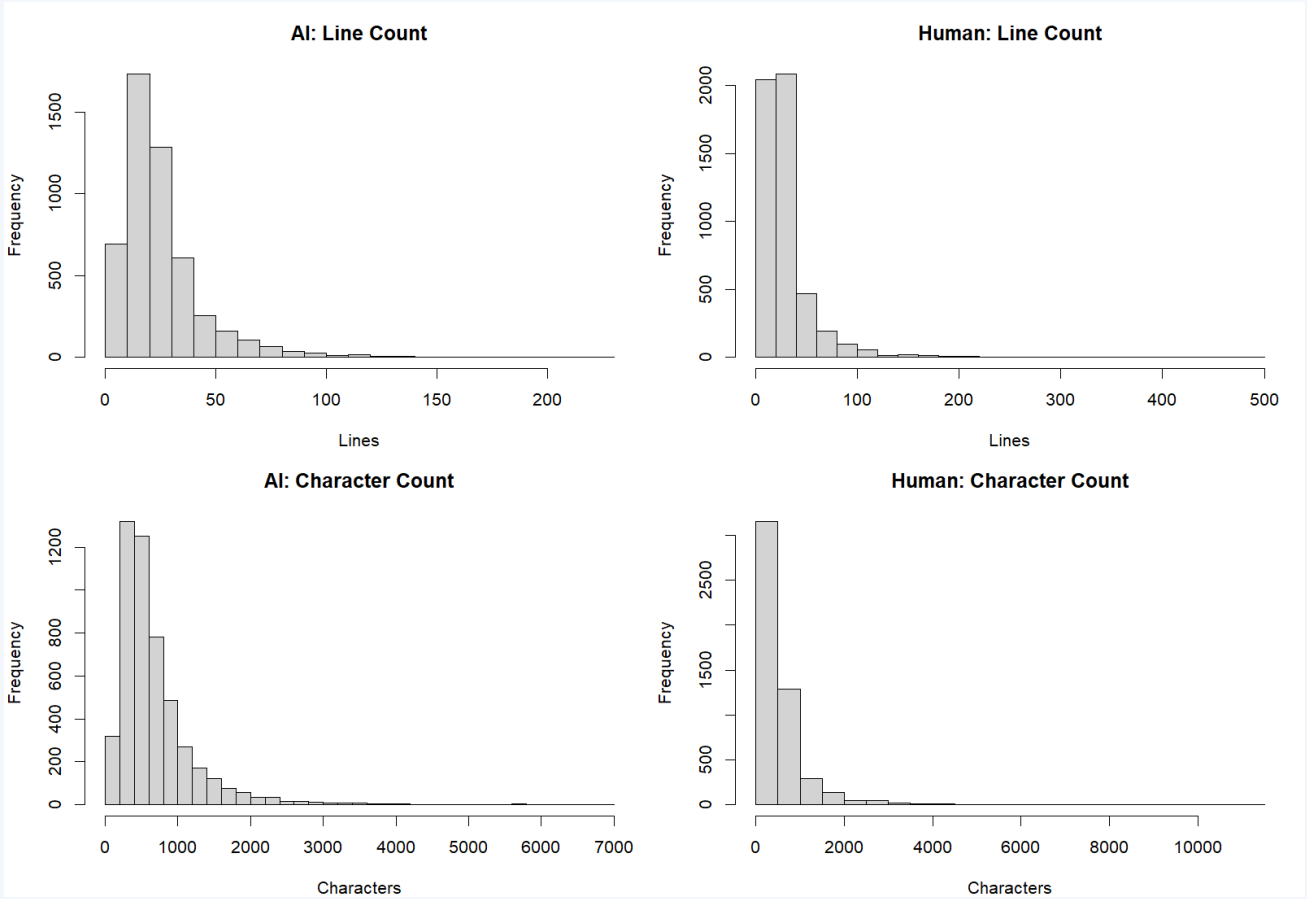
PS C:\Users\akank\OneDrive\Desktop\pasproj> ls


    Directory: C:\Users\akank\OneDrive\Desktop\pasproj


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
dar---l       16-10-2026     08:34                Code_Dataset
dar---l       15-10-2026     23:08                __MACOSX
-a----l       30-10-2026     19:55        9324414 .RData
-a----l       31-10-2026     21:12          13352 .Rhistory
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                                                  2.pptx
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-a----l       16-10-2026     06:56         749045 Review1_Human_vs_AI_Code_Report.pdf


PS C:\Users\akank\OneDrive\Desktop\pasproj>
```

Quality check: what our data can and can’t support



Different source pipelines

Human code came from a repository, AI code was generated. So source task differences may confound origin.

Heuristic comment counter

May miss inline comments or misread strings or block comments. If someone uses our study, do manually annotate the comments for accuracy.

Independence & prevalence

Only 3 problem IDs appeared in both origins, which is less but still full independence is not proven. A 50/50 split is not real-world prevalence.

Interpretation boundary

We make no claims about code quality, correctness, misconduct or individual authorship. This is not a validated detector, but rather a statistical study which can supplement those and further research. Our study shouldn’t be used to make total decisions.

It’s not just that AI code is longer, the pattern is more specific:

01

DENSITY

AI samples had higher median characters per nonblank line in all four languages.

26.1 vs 15.1 in C, 32.2 vs 19.1 in Python.

02

COMMENTS

AI samples were more likely to contain a detectable comment line in every language.

Largest gap: Python +28.5 percentage points.

03

LANGUAGE MATTERS

Raw length patterns vary by language, ANOVA found language by origin interactions.

C++ character count mean: has no Welch significance.

Our contribution is statistically interpretable, not an explicit detector.

SELECTED SOURCES

References and reproducibility

[1] Boukili, G., El Garouani, S., & Riffi, J. (2026). “A dataset for human-written and AI-generated code source classification.” Data in Brief, 65, 112527. https://doi.org/10.1016/j.dib.2026.112527

[2] Demirok, B., Kutlu, M., & Mergen, S. (2026). “MultiAIGCD: A comprehensive dataset for AI generated code detection covering multiple languages, models, prompts, and scenarios.” Neural Computing and Applications, 38, Article 687. https://doi.org/10.1007/s00521-026-12336-0

[3] Gurioli, A., Gabbrielli, M., & Zacchiroli, S. (2024). “Is This You, LLM? Recognizing AI-written Programs with Multilingual Code Stylometry.” arXiv:2412.14611. https://arxiv.org/abs/2412.14611

[4] Demirok, B., & Kutlu, M. (2024). “AIGCodeSet: A New Annotated Dataset for AI Generated Code Detection.” arXiv:2412.16594. https://arxiv.org/abs/2412.16594

Dataset access: https://data.mendeley.com/datasets/kjh95n54f8/1

Reproducibility: you can retain our clean R script, dataset description, output CSVs and plots with the final report.

Thank you!

We’re open to questions & discussion!

Email us at akankshagupta4344@gmail.com.

A statistical comparison of code characteristics across AI-generated and human-written samples.

Akanksha Gupta (Ideation, Module 1,4,5) & Bhargavi Panwar (Module 2&3), Probability & Statistics Semester Project